

ESSIAW CHARLES JNR

Software Developer & Full-Stack Systems Engineer

Accra, Ghana (Remote & Relocation Ready) | +233 53 798 4448 | charlesessiawjnr@gmail.com
Portfolio: kojowallet01.github.io/essiawcharles | GitHub: github.com/kojowallet01 | LinkedIn: [charles-essiaw](#)

PROFESSIONAL SUMMARY

High-velocity Full-Stack Software Developer specializing in resilient web applications, real-time event-driven systems, and clean UI engineering. Proven track record architecting mission-critical production platforms including cloud restaurant POS/KDS systems, national emergency dispatch command software, and cryptographic QR access control. Extensive proficiency in TypeScript, Next.js (App Router), Supabase, PostgreSQL, Python, and Docker with an unwavering engineering commitment to sub-second latency, 100% Core Web Vitals, and modular, testable codebases.

TECHNICAL SKILLS & ARCHITECTURE MATRIX

Languages:	TypeScript, JavaScript (ES6+), Python, SQL (PostgreSQL), HTML5, CSS3, Bash
Frontend Architecture:	Next.js 14/15 (App Router), React, Tailwind CSS, Component Architecture, State Machines, Core Web Vitals
Backend & Realtime:	Node.js, Supabase (Realtime & Auth), RESTful APIs, WebSockets, Python (FastAPI/Flask), Server Actions
Databases & Cloud:	PostgreSQL, Relational Data Modeling, Docker, Vercel, Render, Git/GitHub, Linux Infrastructure
Engineering Disciplines:	Real-Time Event Pipelines, Point of Sale Systems, QR Cryptographic Security, GPS Geolocation, Web Performance

KEY ARCHITECTURAL SOFTWARE PROJECTS

Sweetbite Food POS & Kitchen Display System (KDS) Active Production System

Next.js 14, TypeScript, Supabase Realtime, PostgreSQL, Tailwind CSS

- Real-Time Telemetry:** Architected an end-to-end synchronized cloud Point of Sale (POS) and Kitchen Display System (KDS) for busy operations, replacing physical paper tickets and eliminating kitchen communication bottlenecks.
- Sub-Second Synchronization:** Leveraged Supabase Realtime WebSocket channels with bi-directional state sync to broadcast modifications, and ticket status changes across kitchen stations with sub-500ms latency.
- Kitchen Operations UX:** Designed auditory kitchen alert chime systems, high-contrast touch ticket layouts, automated table-routing, and updates for seamless offline and reconnection recovery.
- Data Integrity:** Modeled relational PostgreSQL schemas with normalized order ticket states, line items, and station-routing tables to guarantee zero order collision during concurrent dining rushes.

Ghana Emergency Response & Dispatch Telemetry System Live Deployment

TypeScript, JavaScript, Python, PostgreSQL, Geolocation API, WebRTC

- Rapid Citizen SOS:** Engineered a mission-critical emergency web application enabling instant geolocation-tagged SOS distress calls and dispatch coordination of emergency response units across Ghana.
- Live Dispatch Telemetry:** Developed live coordinate telemetry with interactive mapping, geofencing, and instantaneous voice-notifying incident verification and response coordination times.
- Offline Fault Tolerance:** Architected high-reliability client-side offline queues and service worker persistence to safely cache emergency requests under intermittent or severed cellular connectivity.
- Command Console:** Implemented administrative dispatch console allowing operators to triage incidents by severity, visualize proximity of emergency units, and update live call statuses.

Patron Housing Access Control & Visitor Management Production Solution

Python, PostgreSQL, Docker, QR Cryptographic Passports, Modern CSS

- Cryptographic Access:** Engineered automated residential access control platform handling digital visitor pass generation, security gate access, and cryptographic QR code validation.
- Infrastructure & Ops:** Built centralized administrative dashboards with real-time entry/exit logs, resident authorization workflows, and deployment on Render cloud infrastructure.
- Tamper-Proof Audit:** Implemented automated security revocation protocols and single-use digital visitor passes, preventing credential reuse and unauthorized residential access.

ARCHITECTURAL METRICS & PERFORMANCE BENCHMARKS

Sub-500ms Broadcast:	Supabase Realtime WebSockets broadcasting concurrent kitchen order updates.
100% Core Web Vitals:	Zero render-blocking scripts, sub-second LCP, and lightweight vanilla architecture.
Zero Data Collision:	PostgreSQL strict transactional safety, parameterized queries, and RLS policies.
Offline Resilience:	Graceful degradation, client-side event queues, and instant reconnection re-sync.

ENTERPRISE COMMERCIAL PLATFORMS & DEPLOYMENTS

BizConnect Technologies Enterprise Portal Client Production

Modern JavaScript, Accessible UI/UX, Performance Optimization, REST APIs

- **Zero-Bloat Engineering:** Engineered enterprise corporate web platform for BizConnect Technologies featuring zero external bundle bloat, clean layouts, and a perfect 100% Core Web Vitals audit.
- **Enterprise Accessibility:** Implemented WCAG 2.1 AA accessible typography, keyboard navigation, and semantic HTML structure for universal accessibility across enterprise clients.
- **Lead Generation:** Built modular component system and client-side contact inquiries API integration, driving a 35% increase in verified enterprise leads.

Kelrose Tours Travel & Itinerary Booking Platform Client Production

Dynamic JavaScript, Modern Responsive CSS, Tour Catalog, SEO Architecture

- **Tour Architecture:** Designed and deployed full-featured travel platform for Kelrose Tours featuring interactive tour itinerary catalogs, destination guides, and conversion funnels.
- **Mobile Optimization:** Optimized responsive layout, image delivery pipelines, and metadata indexing, enhancing organic search visibility and user retention.
- **Booking Operations:** Configured automated inquiry workflows routing customer booking requests directly to operations staff via communication channels.

PROFESSIONAL ENGINEERING EXPERIENCE & MILESTONES

Full-Stack Systems Engineer & Independent Developer 2024 - Present

Autonomous Software Engineering & Client Solutions - Accra, Ghana

- **Full-Lifecycle Engineering:** Lead the complete software engineering lifecycle for high-stakes web systems across hospitality, municipal, and commercial enterprise sectors.
- **Database & Cloud Delivery:** Architect robust database models in PostgreSQL, implement secure role-based access control, w event-driven services, and manage continuous deployment on cloud infrastructure.
- **Client Excellence:** Maintain a 5.0 Google client satisfaction rating across commercial engagements through meticulous requirement management, disciplined delivery timelines, and proactive communication.

Commercial Web Engineer & Client Consultant 2023 - 2024

Freelance & Contract Engagements

- **Frontend Modernization:** Consulted with SME business owners to modernize legacy web applications, migrate outdated static pages to modern stacks, and optimize conversion funnels.
- **Performance Engineering:** Implemented performance audits and asset optimization techniques, reducing page load times by over 40% and sub-second Largest Contentful Paint (LCP).
- **Cross-Device Resilience:** Delivered responsive UI components adhering strictly to mobile-first standards, testing across multi-device and legacy browser runtimes.

Autonomous Software Apprenticeship & Foundations 2022 - 2023

Intensive Systems Engineering & Deliberate Practice (1,500+ Hours)

- **Theoretical Foundations:** Completed rigorous self-directed engineering curriculum covering algorithmic complexity, asynchronous conc protocols (HTTP/REST/WebSockets), and relational database modeling.
- **AI-Augmented Engineering:** Pioneered AI-augmented development workflows, pairing with cutting-edge agentic tools to prototype, refa stress-test codebases at accelerated speed.
- **Community & Practice:** Authored reproducible open-source web components and modular architectural blueprints shared with local communities in Accra.

EDUCATION, CERTIFICATIONS & CONTINUOUS MASTERY

Software Engineering & Systems Development Accra, Ghana

Autonomous Curricula, Open-Source Contributions & Professional Practice

- Continuous professional mastery in Next.js App Router, Advanced TypeScript Typing, PostgreSQL Query Optimization, and Cloud Native Deployments.
- Committed to perpetual engineering refinement through daily deliberate practice, code review, architectural post-mortems, and modern web specifications.

CORE ENGINEERING METHODOLOGIES & ARCHITECTURAL STANDARDS

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| Resilient System Design: | Offline-first thinking, fallback event queues, optimistic UI state updates, and graceful network degradation. |
| Performance Discipline: | Zero unnecessary dependencies, sub-second LCP, minimal DOM reflows, and rigorous Core Web Vitals adherence. |
| Code Quality & Delivery: | Modular domain separation, type safety with TypeScript, clean git branching workflows, and rapid automated deployment. |